

TermPlanMT

Graph-Grounded Terminology Planning for Regulatory Medical Document Translation

Machine Translation Project Proposal • Darragh Kerins

1. Introduction

Neural MT produces fluent output but does not enforce consistent terminology. In regulatory medical documents the same term must appear identically across hundreds of segments, and mistranslation can have direct patient-safety consequences. Three failure modes are common:

- **Term drift:** the same source term is translated differently across a document.
- **Concept flattening:** a specific term is replaced by an over-general hypernym.
- **Coverage gaps:** domain-specific terms are omitted or rendered inconsistently.

The EMA publishes Summaries of Product Characteristics (SmPCs) for every approved drug across all 24 EU languages. Each SmPC has a certified human English translation, making this corpus ideal for MT evaluation. MedDRA (the Medical Dictionary for Regulatory Activities) provides a validated five-level ontology covering the adverse events, indications, and drug classes found in these documents. Its hierarchy runs from broad to specific:

Level	Tier	Example
L1	System Organ Class (SOC)	Respiratory disorders
L2	High Level Group Term (HLGT)	Pulmonary conditions
L3	High Level Term (HLT)	Pneumonitis
L4	Preferred Term (PT)	Pembrolizumab-related pneumonitis
L5	Lowest Level Term (LLT)	Immune-mediated pneumonitis

TermPlanMT connects these two resources: it reads a source French SmPC, grounds its terminology in a persistent Neo4j concept graph built from MedDRA, and enforces that each translated term lands at the same hierarchy level as its source.

Scope. French–English SmPCs for checkpoint inhibitors (pembrolizumab, nivolumab, atezolizumab, ipilimumab, durvalumab). This drug class is chosen because its SmPCs span the full MedDRA hierarchy across every System Organ Class, giving the densest possible terminology coverage.

Why it matters. Indications, contraindications, and special warnings in a SmPC carry regulatory force across every EU jurisdiction. A term translated at the wrong ontology level (e.g. *pneumonitis* instead of *immune-mediated pneumonitis*) shifts clinical meaning, introduces cross-lingual inconsistencies in pharmacovigilance reporting, and may not satisfy EMA linguistic review. The persistent Neo4j graph is also a reusable by-product: grounded terminology accumulates across all processed SmPCs and can be queried independently of the MT pipeline.

2. Research Questions

1. Does graph-based terminology planning improve TCR over zero-shot MT and flat glossary lookup?
2. Does structured ontology grounding improve hierarchy preservation, as measured by HTM?
3. Does a persistent Neo4j graph reduce grounding latency and improve CCR across a multi-document corpus?

4. Can a long-context LLM with full-document awareness (e.g. Qwen2.5-1M, BioMistral) match or exceed graph-based planning for term consistency, without ontology grounding?

3. Related Work

Constrained decoding. Hokamp & Liu [1] introduced grid beam search for hard lexical constraints; Post & Vilar [2] extended this with dynamic beam allocation (used here). Neither approach addresses *which* constraint to select when multiple ontologically related candidates exist. That is what the planning step solves.

Terminology in NMT. Dinu et al. [3] and Bergmanis & Pinnis [4] employ flat glossaries with no concept structure. TermPlanMT selects constraints from a hierarchy, not a list.

Document-level MT. Maruf et al. [5] improve lexical coherence via selective attention, but without ontological grounding or hard enforcement.

Biomedical NER. Martin et al. [6] provide CamemBERT for French; English-only biomedical models are not appropriate for the source language here.

MT evaluation. BLEU [7], chrF [8], and COMET [9] all measure fluency but none captures hierarchy preservation. This is why we introduce HTM.

Long-context LLMs. Recent open-source models such as Qwen2.5-1M [14] and BioMistral [15] offer context windows of up to one million tokens, which is in principle large enough to process an entire SmPC in a single pass (the French source SmPC at 368 pages and $\approx 129,700$ words estimates to $\approx 168,600$ tokens (at ~ 1.3 tokens per word), well within this limit). This raises a direct question for TermPlanMT: can raw long-context awareness alone solve the consistency problem, without explicit ontology grounding? We include a long-context LLM as an additional baseline (B4) to answer this.

4. Data

4.1 Primary Evaluation Document

The primary evaluation document is the **KEYTRUDA (pembrolizumab)** SmPC, available in French and English directly from the EMA product database [13]:

<https://www.ema.europa.eu/en/medicines/human/EPAR/keytruda>

This document is selected because it is long (FR source: 368 pages, $\approx 129,700$ words; EN reference: 341 pages). The French source is the longer version, as is typical for FR/EN regulatory translation, and yields $\approx 2,600$ segments, officially translated, and exhibits all three failure modes at scale:

- **Term drift** (Section 4.8): *pneumopathie immuno-médiée* appears in over 30 segments across the undesirable effects section. Zero-shot MT produces at least three distinct English renderings (*immune pneumopathy*, *immune-mediated pneumonitis*, *immunological lung disease*) with no consistency mechanism.
- **Concept flattening** (Section 4.8): LLT-level terms such as *pneumopathie interstitielle immuno-médiée* (L5) are frequently collapsed to *pneumonitis* (L3/L4), a valid but over-general rendering two levels too high, losing the specificity that determines clinical management.
- **Coverage gaps** (Section 4.4): Multi-token adverse event names spanning clause boundaries (e.g. *syndrome de Steven-Johnson d'origine immunitaire*) fall outside standard NER and are likely missed, leaving them unconstrained.

Predicted failure points. Three translation problems are anticipated for this document specifically:

1. **NER boundary errors.** The pembrolizumab SmPC contains long nominal phrases fusing a drug name, immune qualifier, and anatomical location (e.g. *colite immuno-médiée de grade 3*). CamemBERT-NER is likely to tag the drug name correctly but miss or split the full compound adverse event expression.
2. **MedDRA lookup failures for novel combinations.** Checkpoint inhibitor immune-related adverse events are a relatively recent clinical category. Some LLT entries may be absent or misaligned in the MedDRA version available, causing fallback to unconstrained translation precisely for the most domain-specific terms.
3. **Constraint conflicts in dense segments.** Section 4.8 contains segments with five or more pharmaceutical terms. Simultaneous enforcement of that many constraints significantly increases the risk of beam starvation and fluency loss despite the beam expansion mechanism.

4.2 Supporting Evaluation Documents

Nine further SmPCs (two per drug: nivolumab, atezolizumab, ipilimumab, durvalumab) extend the evaluation across mechanisms and indication sets, providing cross-document TCR measurement.

4.3 Training Corpus

The EMEA parallel corpus [11], a large collection of EMA documents in 22 languages, is filtered to French–English SmPC segments containing at least one MedDRA Preferred Term, yielding $\approx 15,000$ – $20,000$ sentence pairs. Split: 80/10/10 train/dev/test; all evaluation SmPCs are held out entirely.

4.4 Document Processing and Context Windows

Traditional NMT models handle 3–5 pages per pass. The French Keytruda SmPC is 368 pages, so naive segment-level processing loses cross-document context. The pipeline processes the document section by section (SmPC sections 1–16 per EMA structure), keeping each chunk within the NMT token limit. The Neo4j graph maintains terminology consistency globally across all sections.

Long-context LLMs change this picture significantly. Dense regulatory text runs at approximately 400 tokens per page, so the Keytruda SmPC falls well within the context limits of current models:

System	Context (tokens)	Keytruda fit?	Requires chunking?
Traditional NMT	$\sim 1,000$	No ($\approx 62,000$ needed)	Yes
NLLB-200 (baseline)	$\sim 1,024$	No	Yes
BioMistral [15]	$\sim 32,000$	No ($\approx 4\times$ too small)	Yes
Qwen2.5-1M [14]	1,000,000	Yes ($\approx 169,000$ tokens)	No

Table 1: Context window comparison. The French Keytruda SmPC (source language) at 368 pages and $\approx 129,700$ words is estimated at $\approx 169,000$ tokens, fitting comfortably in a 1M-token window. The English reference is 341 pages ($\approx 136,000$ tokens). BioMistral (32k context) still requires chunking for both.

This motivates the B4 ablation: if a model can see the entire document at once, does it achieve consistent terminology without ontology grounding? The hypothesis is that long-context awareness improves fluency coherence but does not enforce correct MedDRA specificity levels, which requires explicit grounding.

5. System Architecture

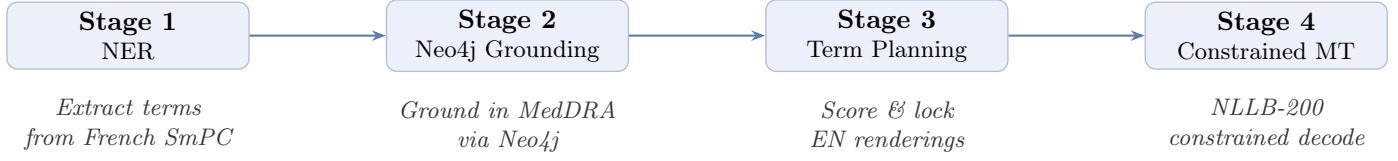


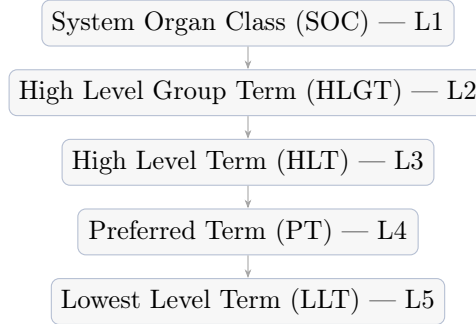
Figure 1: TermPlanMT four-stage pipeline.

5.1 Stage 1 — Term Extraction

A CamemBERT-based NER model [6] fine-tuned on the QUAERO French Medical corpus [10] processes the full source SmPC in a single pass, tagging drug names, drug classes, adverse events, anatomical terms, dosage forms, and quantities. No additional fine-tuning is required.

5.2 Stage 2 — MedDRA Grounding and Persistent Neo4j Graph

Each extracted term is looked up in MedDRA via its REST API. Successful lookups return a concept identifier, preferred English label, hierarchical level, and parent/child identifiers. MedDRA’s five levels map directly onto Neo4j node and relationship types:



The Neo4j instance is initialised once and accumulates terminology across all SmPCs. Terms already grounded from a prior document are retrieved directly via Cypher, skipping the API call. Grounding latency therefore decreases as the corpus grows.

```

from neo4j import GraphDatabase
driver = GraphDatabase.driver("bolt://localhost:7687", auth=("neo4j", "password"))

def ground_term(tx, fr_term, concept_id, en_label, level, parent_id):
    tx.run("""
        MERGE (c:Concept {id: $concept_id})
        SET c.fr_label = $fr_term, c.en_label = $en_label, c.level = $level
        WITH c MATCH (p:Concept {id: $parent_id})
        MERGE (p)-[:BROADER_THAN]->(c)
    """, concept_id=concept_id, fr_term=fr_term,
        en_label=en_label, level=level, parent_id=parent_id)

def query_term(tx, fr_term):
    return tx.run("""
        MATCH (c:Concept {fr_label: $fr_term})
        RETURN c.en_label AS label, c.level AS level, c.id AS id
    """, fr_term=fr_term).single()
  
```

Coverage is tracked with $CCR = |\text{grounded}|/|\text{extracted}|$, reported per SmPC and cumulatively.

5.3 Stage 3 — Terminology Planning

When MedDRA returns multiple valid English renderings, each candidate c is scored:

$$\text{Score}(c) = \alpha \cdot \text{Context}(c) + \beta \cdot \text{Neighbour}(c) - \gamma \cdot \text{HierarchyPenalty}(c) \quad (1)$$

- **Context**(c): cosine similarity to the local sentence (`paraphrase-multilingual-mpnet-base-v2`).
- **Neighbour**(c): mean similarity to the selected renderings of graph-adjacent concepts (parents, children, siblings), retrieved via Cypher.
- **HierarchyPenalty**(c) = $|d_{\text{MedDRA}} - d_{\text{source}}|$: penalises candidates at the wrong specificity level.

Weights $\alpha, \beta, \gamma \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ are grid-searched on the dev set and validated jointly on TCR and COMET to prevent overfitting. A global consistency pass locks the top-scoring rendering for each term before any segment is translated.

5.4 Stage 4 — Constrained Decoding

NLLB-200 [12] (`fra_Latn→eng_Latn`) is used as the translation model. Dynamic Beam Allocation [2] partitions the beam by constraint-satisfaction state, guaranteeing a fully-constrained hypothesis always reaches the final beam. Beam size defaults to 5, expanded to 10 for segments with three or more simultaneous constraints.

6. Gold-Standard Term Annotation

A manual FR→EN term mapping is built for all unique medical terms in the evaluation SmPCs, with MedDRA levels verified against the official MedDRA browser. Sample entries from the pembrolizumab SmPC:

French	English (gold)	L	Tier
Antinéoplasiques	Antineoplastic agents	1	SOC
Inhibiteurs de point de contrôle	Checkpoint inhibitors	2	HLGT
Inhibiteur PD-1	PD-1 inhibitor	3	HLT
Pembrolizumab	Pembrolizumab	4	PT
Pneumopathie immuno-médiée	Immune-mediated pneumonitis	5	LLT

Table 2: Sample gold-standard mappings from the pembrolizumab SmPC, illustrating the full five-level MedDRA hierarchy.

7. Evaluation

7.1 Baselines

ID	System	Terminology	Hierarchy-aware
B1	NLLB-200 zero-shot	None	No
B2	NLLB-200 fine-tuned (EMEA)	None	No
B3	NLLB-200 + flat MedDRA lookup	Flat	No
B4	Qwen2.5-1M zero-shot (full doc.) [14]	None	No
Prop.	TermPlanMT	Graph	Yes

Table 3: B3 isolates the planning step’s contribution over flat lookup. B4 tests whether full-document long-context awareness alone achieves consistency without ontology grounding.

7.2 Metrics

Metric	Measures	Tool
BLEU	n -gram fluency	sacrebleu
chrF	Character F -score	sacrebleu
COMET	Neural translation quality	unbabel-comet
TCR	Term translation consistency	Custom
HTM	Hierarchy-aware accuracy (novel)	Custom
CCR	Concept grounding coverage	Custom

Table 4: Evaluation metrics.

7.3 HTM — Hierarchy-Aware Terminology Metric

HTM measures how far a translated term has drifted from its source position in the MedDRA hierarchy. Each MT output term is grounded in Neo4j to recover its level $\hat{d}_i$, then compared against the source level d_i from the Stage 2 graph:

$$\text{HTM} = \frac{1}{N} \sum_{i=1}^N s_i, \quad s_i = \begin{cases} 1.0 & \hat{d}_i = d_i \quad (\text{exact level match}) \\ 0.5 & \hat{d}_i \neq d_i, \text{ same branch (wrong specificity)} \\ 0.0 & \text{different branch or ungrounded} \end{cases} \quad (2)$$

where N is the total number of successfully grounded terms in the evaluation set. TCR is computed at term level within each segment (counting unique source terms rendered consistently) and CCR at document level; both are then aggregated corpus-wide. HTM is additionally reported per SmPC section (4.1, 4.3, 4.4, 4.8) to reveal whether errors concentrate in legally critical zones.

7.4 HTM Taxonomy Example

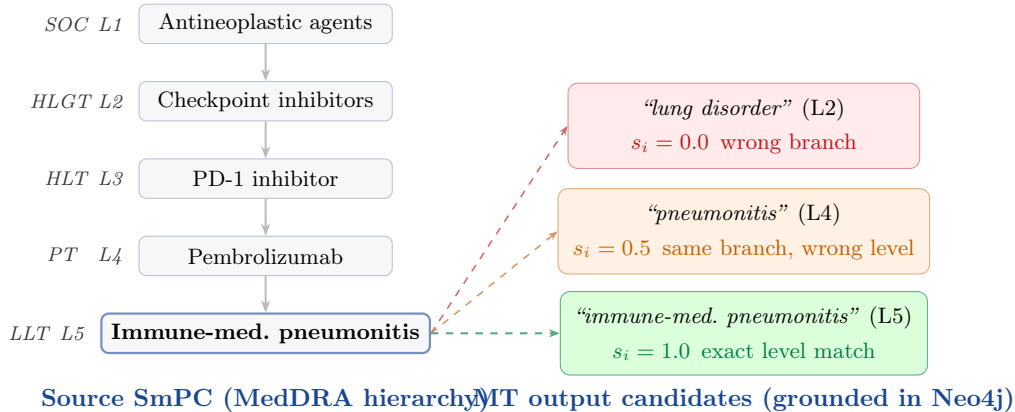


Figure 2: HTM scoring for *pneumopathie immuno-médiationnée* (source: LLT, L5). Each MT output term is grounded in Neo4j and its MedDRA level compared against the source. Score and reason are shown inline in each candidate box.

7.5 Error Analysis

Twenty randomly sampled segments per system are annotated for: term drift (≥ 2 renderings of the same term), concept flattening (over-general hypernym substituted for the correct term), and fluency loss (grammatically correct but unnatural output). Results are broken down by SmPC section.

7.6 Expected Results

Success thresholds: TCR $\geq +5$ pp over B3; HTM $\geq +0.05$ over B3; BLEU/COMET within 1 point of B2.

Metric	Type	B1	B2	B3	TermPlanMT
TCR	Terminology	$\sim 68\%$	$\sim 75\%$	$\sim 82\%$	87–91%
HTM	Terminology	~ 0.60	~ 0.65	~ 0.70	0.76–0.81
BLEU	Fluency	~ 36.0	~ 39.5	~ 39.0	39.5–41.5
COMET	Fluency	~ 0.81	~ 0.83	~ 0.83	0.83–0.85

Table 5: Projected results. Baselines reflect NLLB-200 performance on biomedical text [12], domain adaptation gains [3], and the flat-glossary TCR gap [4].

8. Contributions

1. **Pipeline.** French NER $\rightarrow$ MedDRA grounding via persistent Neo4j $\rightarrow$ graph-based planning $\rightarrow$ constrained decoding.
2. **HTM.** A novel graph-comparison metric measuring MedDRA level drift between source and translated terms.
3. **Ablation.** Graph-based vs. flat-lookup comparison isolating the hierarchy’s contribution.
4. **Knowledge graph.** A reusable Neo4j pharmaceutical terminology graph, accumulated across all processed SmPCs.

9. Limitations

- **MedDRA coverage.** Terms absent from MedDRA fall back to unconstrained translation; CCR quantifies this gap.
- **NER recall.** Missed source terms cannot be grounded or constrained; pipeline performance is bounded by Stage 1 recall.
- **Fluency trade-off.** Hard constraints can produce unnatural output in dense segments; beam widening mitigates but does not eliminate this.
- **Scope.** Checkpoint inhibitors, French–English only; extension to further drug classes and language pairs is future work.

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